# Unit 5: Programming & Lab Practical

This final unit brings all the previous concepts to life. We move from logic on paper to actual execution using two of the most widely used languages in computing: **C** (a structured, high-level language) and **Python** (an interpreted, high-level language).

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## 5.1 Language Overview: C vs. Python

While both can solve the same problems, they have different "philosophies":

- **C:** Known for speed and efficiency. It is closer to the hardware and requires explicit data typing.
- **Python:** Known for readability and simplicity. It allows for faster development with fewer lines of code.

## 5.2 Basic Syntax Comparison

Let's look at how the core statements discussed in Unit 4 are written in both languages.

### 5.2.1 Variables and Assignment

- **In C:** You must declare the type of data (e.g., int for integer).
  int age = 20;
- **In Python:** The type is determined automatically.
  age = 20

### 5.2.2 Input and Output (I/O)

| Task | C Syntax | Python Syntax |
|---|---|---|
| **Output** | printf("Hello World"); | print("Hello World") |
| **Input** | scanf("%d", &age); | age = int(input("Enter age: ")) |

**5.2.3 Conditional Statements (The if Logic)**

- **C Example:**
- C

```
if (score >= 40) {
    printf("Pass");
} else {
    printf("Fail");
}
```

- 
- 
- **Python Example:**

Python

```
  if score >= 40:
      print("Pass")
   else:
      print("Fail")
  ```
```

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### 5.3 Lab Work: Practical Exercises

The lab component is designed to build muscle memory for problem-solving. Students will use tools to bridge the gap between design and code.

#### Activity 1: Flowcharting & Pseudocode

* **Tools:** Use simple tools (like Lucidchart, Draw.io, or even pen and paper) to draw flowcharts for everyday processes (e.g., "Making a cup of tea" or "Withdrawing money from an ATM").
* **Task:** Write pseudocode for a program that calculates the average of three numbers.

#### Activity 2: Coding Simple Programs

Students will implement the following logic in either C or Python:

1. **The Calculator:** Accept two numbers and an operator (+, -, *, /) and display the result.
2. **The Even/Odd Checker:** Take an integer input and determine if it is divisible by 2.
```

3. **The Grade Calculator:** Use nested `if` statements to assign grades (A, B, C, D, F) based on a student's score.

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### 5.4 Best Practices in the Lab
* **Indentation:** Crucial in Python for logic; crucial in C for readability.
* **Comments:** Use `#` in Python or `//` in C to explain what your code is doing.
* **Incremental Development:** Write a few lines, test them, and then move on. Never write 50 lines of code at once without testing.

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### Summary of COS 102
We have moved from **identifying a problem** (Unit 1), to choosing a **technique** (Unit 2), **designing the logic** (Unit 3), **preparing for implementation** (Unit 4), and finally **writing the code** (Unit 5).

**Course Wrap-up:**
The goal of this course is not just to learn a language, but to learn how to **think** like a problem solver. A language is just a tool; the logic is the solution.

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**This completes the lecture notes for COS 102.** Would you like me to generate a sample **continuous assessment (CA) test** or a **final exam** based on these five units to round out the material?